ECON 5321 Group Project Report

The Impact of Expert Reviews on Wine Demand: Evidence from the Swedish Market

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1 Overview and Empirical Framework

1.1 Overview

This article uses detailed data from the Swedish wine market to examine whether expert reviews have a causal impact on wine demand. The authors focus on whether reviews influence consumers' purchasing behaviour, and how the effects differ depending on the type of review, such as positive, neutral, or negative. A key challenge addressed in the paper is distinguishing between correlation and causation, since wines with higher ratings may already have better quality and therefore naturally attract higher demand.

To examine this issue, the authors used data covering five years (2002-2007), including information on sales, advertising spending and expert reviews from six major Swedish media organisations. This dataset enabled them to examine the changes in demand over both short-term and long-term periods. By controlling for wine characteristics (such as price, vintage, and product identity) as well as changes in demand across different market segments, the model provides a reasonable estimate of what sales might have looked like without the influence of reviews.

An important strength of the paper lies in the institutional setting of the Swedish market. Wine retailing is controlled by a government monopoly (Systembolaget), which does not allow retailers to adjust prices, shelf placement, or promotional activities in response to reviews. This removes many potential confounding factors that would otherwise make causal identification difficult. In addition, the paper exploits the annual Allt om Mat wine tasting issue, where a large number of wines are reviewed at predetermined dates, providing exogenous variation in review timing.

The paper finds that expert reviews have a clear impact on wine demand. Positive reviews can increase sales by about 6% at their peak, and this effect can last for more than 20 weeks. Neutral reviews only have a small effect, while negative reviews do not seem to change demand much. This shows that expert assessments can influence consumer behavior. This influence can be observed shortly after the assessment is released or persist over a longer period.

1.2 Research Objective and Empirical Approach

This section outlines the research objective and the empirical strategy used to identify the causal impact of expert reviews on wine demand.This paper aims to explore whether expert reviews have a causal impact on wine demand, and whether the nature of the reviews (positive, neutral, or negative) has different effects on sales. The core issue is distinguishing between causation and simple correlation, as wines receiving positive reviews may inherently possess higher quality or have greater demand.

To address this issue, the author conducted an in-depth analysis of the weekly data of the Swedish wine market from 2002 to 2007. Furthermore, they combined information from six major media sources regarding sales, advertising expenditures, and expert evaluations. They estimated a distributed lag model, where the dependent variable was the logarithm of weekly wine sales, and the key explanatory variables were whether the wine was evaluated and whether the evaluation was positive or negative. The model included fixed effects for wine $\times$ year $\times$ price and week $\times$ category to control for product characteristics and the changing demand patterns over time.

The author further points out that the institutional environment in Sweden is more conducive to explaining the causal relationship. The sale of wine is handled by a local retailer that is monopolised and operated by the government. This retailer does not adjust prices, positions, or conduct promotional activities based on evaluation results. Additionally, they utilise the annual Allt om Mat wine tasting special issue, in which almost all wines are evaluated at pre-determined times, to provide exogenous variation in the timing of evaluations.

1.3 Main Findings

The main conclusion of this paper is that positive evaluations will significantly increase people's demand for wine. The effect produced by positive evaluations is the highest in the first week after the evaluation is released, leading to an increase in demand of approximately 6%. And this effect remains statistically significant for the next 20-plus weeks.

The baseline results show that a single assessment itself will increase sales by about 1.2% in the same week, and its effect can last for several weeks. Positive evaluations can bring about sustained and increasing demand for sales, while neutral evaluations have little impact. In contrast, negative evaluations have almost no effect and thus do not have a significant impact.

From an economic perspective, a single positive review can increase sales by 4,418 units. This is equivalent to half of the average weekly sales volume, amounting to approximately 440,000 Swedish kronor in additional revenue. The study also found that the effect of advertising was weaker than the effect of evaluation, and also shorter in duration.

1.4 Identification Challenges and Solutions

A key challenge in estimating this relationship is addressing potential endogeneity concerns. One of the key endogeneity issues in this paper is the possibility that expert evaluations may be related to unobserved demand shocks. Wines that have become increasingly popular are more likely to be evaluated, which creates a reverse causal relationship and leads to biased estimates of the evaluation effect.

To address this issue, the authors introduce fixed effects that control for seasonal demand variations within different wine categories. In addition, they include lead values of the evaluation variables to test for pre-existing trends and find that these coefficients are generally not significant, indicating that the evaluations are not driven by prior growth in demand.

The second issue is that the timing of evaluations may not be exogenous. If evaluations are strategically arranged or influenced by unobserved factors, the estimated effects may still be biased. The authors address this problem by using the annual Allt om Mat wine tasting issue, in which almost all wines are evaluated at predetermined dates. Since this timing is exogenous to individual wine demand, the similarity of the results provides strong support for a causal interpretation.

The final issue is that the responses of retailers or wholesalers may confound the estimated effects. If the store adjusts prices, shelf placements, or promotional strategies based on customer reviews, the estimated impact of the reviews may include both direct and indirect effects. This article argues that in Sweden, this issue is not serious because monopolistic retailers are not allowed to flexibly adjust prices, shelf placements, or promotional strategies. This helps to isolate the impact of reviews on demand.

2 Discussion of Expert Reviews and Demand

2.1 Importance of Expert Reviews for Consumers

Expert reviews give consumers a glimpse of what the product or experience is without the need to purchase it themselves, thereby reducing information asymmetry. The information tends to be credible, independent and often of high quality as the reviewers do this for a living. This is especially relevant in markets such as wine, where consumers have a lot of uncertainty about which product is considered good.

First, wine is an "experience" good, implying that its true value is in its experience and can only be determined after consumption. Price, brand or country of origin are the observable characteristics that don't really reveal how much enjoyment the wine brings to someone. Expert reviews provide consumers with informed evaluations of a product's quality based on prior consumption, which in this case is wine. They allow consumers to make informed choices and select a product that better matches their preferences. This allows consumers to rely on informed third-party evaluations rather than engaging in a costly trial-and-error.

Second, expert reviews are considered more influential because they are perceived as more credible than other sources of information, such as advertisements. Advertising is done by companies that have a direct financial interest in the company that produced the goods. This causes the consumers to automatically discredit or to mistrust the information provided by advertising agencies. In contrast, Expert reviewers are seen as independent agents that have their own dignity and reputation to uphold. They also do not have any direct financial interest in the sales of the products that they evaluate. This neutrality increases the perceived value of their reviews.

Third, expert reviews provide "soft information" about product quality. Some of them can be objectively measured, such as alcohol content or the region of origin of the wine. Other aspects, such as taste, enjoyment, or which occasion the wine would go better for, are subjective and very hard to quantify and even harder to verify. Expert reviews still try to translate them into easily accessible signals, such as ratings or scores. This simplifies decision-making for customers and reduces search costs.

In addition, a large number of product alternatives causes consumers to suffer from information overload. As noted in the paper, consumers often face hundreds of available options, which is an overwhelming experience. Expert reviews act as a sort of filtering mechanism that simplifies decision-making.

Finally, expert reviews can influence demand in a way similar to informative advertising but often in a more effective manner. The results in the paper show that favourable reviews lead to a significant increase in demand for a long period of time (over 20 weeks, by approximately 6%). As a result, expert reviews play a key role in influencing consumer demand by reducing uncertainty about product quality.

2.2 Market effects of reviews

The authors recognise that if there is an increase in demand for a wine, it comes at the expense of another wine. This is called the crowding-out or substitution effect. To further examine this phenomenon, they extend their baseline empirical model to account for the impact of reviews on similar competing products.

First, they define competing wines as wines within the same taste segment. These taste segments are based on categories used by the retailer (Systembolaget), combining characteristics such as colour (red/white/sparkling), price category, and type of container. This is important because consumers are most likely to substitute between wines that are close alternatives.

They then extend their regression model by including additional variables that capture:

1. The number of competing wines receiving good reviews
2. The number of competing wines receiving bad reviews
3. The total number of competing reviews
4. The total advertising expenditure of competing wines

These variables are also included with leads and lags that capture both immediate and delayed effects on demand.

The results showed us that there is some evidence in the substitution effect, though it's relatively small in comparison. In particular:

1. Good reviews of competing reviews tend to decrease the demand for a given wine, as consumers automatically shift toward the positively reviewed alternatives.
2. Bad reviews of competing wines tend to increase demand, as consumers substitute away from poorly reviewed wines towards others.

However, the authors found that the cross-effects are very small and negligible compared to the effects above. This suggests that the primary effect of reviews is on the reviewed wines' own demand rather than redistributing the demand across competitors and causing serious changes across the market landscape.

To further check the influence on the overall market, the author performed a counterfactual exercise in which they compared predicted sales with and without reviews. They found that total market demand was much higher when reviews were present. This indicates that reviews had a market-expanding effect beyond just redistributing demand across wines.

In summary, the authors take crowding out into account by explicitly incorporating competing reviews into their model. Their findings suggest that while substitution between the wines exists, it is limited. Expert reviews act to increase demand for the reviewed product, with only a modest effect on its competitors.

2.3 Product differentiation

The findings of this paper suggest that the wine market exhibits both vertical and horizontal product differentiation, but with a strong inclination for vertical differentiation.

First, the significant increase in demand following a favourable (good) review indicates the presence of vertical differentiation. Vertical differentiation refers to differences in objective or perceived quality, where all consumers would generally prefer higher-quality products. In this paper, a good review signals higher quality, and the fact that demand increases substantially over time. This shows that consumers respond positively to improvements in perceived quality.

Second, higher-priced wines respond more strongly to good reviews. This further supports the importance of vertical differentiation. As we know, higher-priced wines are typically associated with much greater quality variation, and consumers appear to rely more on expert information when evaluating these products. This suggests that quality differences matter quite a bit and that expert reviews help consumers identify superior options.

However, these results also point to the presence of horizontal differentiation, where consumer preferences are different even if the quality is similar. In particular:

1. Neutral reviews still have a small positive effect on demand
2. Negative reviews do not significantly reduce demand.

This suggests that, despite the fact that there are wines that are not that highly rated. They may still appeal to certain types of consumers due to differences in taste, preferences, or occasions of consumption. In a purely vertically differentiated market, negative reviews would be expected to reduce demand significantly, which is not what is observed here.

Therefore, the evidence implies that while quality differences (vertical differentiation) are important, consumer preferences are also heterogeneous (horizontal differentiation). Wine consumers do not all agree on what constitutes the "best" wine. Some may continue to purchase lower-rated wines just because they match their preference more.

3 Extension:Diminishing Returns to Expert Reviews

Building on the findings of the paper, this section proposes an extension to examine whether the impact of expert reviews diminishes with repeated exposure. Based on the relevant research, we propose a new question: Do repeated reviews exhibit diminishing returns in the wine market?

3.1 Motivation and Relevance

This question is relevant because it concerns the topic of wine and consumer demand. In this paper, Friberg and Grönqvist (2012) show that favourable expert reviews significantly increase wine sales in Sweden. They also find that neutral reviews have small positive effects, and bad reviews have little effect on demand. These findings suggest that expert reviews are an important determinant of wine demand in this market. According to the results, the new question asks whether the same informational mechanism weakens with repetition. If consumers already know a wine well because it has been reviewed before, an additional review may provide less new information. Therefore, it may generate a smaller increase in demand.

3.2 Contribution to the Literature

This question is innovative because the existing literature mainly studies whether expert reviews affect demand, rather than whether the marginal effect of an additional review declines with repeated exposure. This paper by Friberg and Grönqvist (2012) mainly studies whether expert reviews affect wine demand, and whether good reviews matter more than bad reviews. However, it does not study whether later reviews of the same wine are less effective than earlier ones. In their framework, reviews are considered as contemporary and lagged effects. It is not considered a cumulative information process, where the value of another review can decrease over time.

This question is also not answered in other papers. Hilger et al. (2011) study expert opinion in the retail wine market using a field experiment. They show that expert score labels displayed in stores affect consumer purchases. However, their analysis focuses on the information at the time of purchase rather than repeated expert media reviews over time.

Instead of asking whether reviews matter on average, it asks whether repeated reviews matter less. It changes the focus from the average effect of reviews to the information accumulation dynamics. Related research on consumer information suggests that additional reviews may diminish marginal informational value if consumers have already acquired substantial information. Zheng et al. (2024) show in the context of online reviews that additional reviews can diminish marginal effects on consumers' information acquisition and purchase decisions. Though this study is not about wine expert media reviews, it gives support to a hypothesis that repeated expert reviews may also diminish returns in retail wine demand.

3.3 Practical and Economic Implications

This question is useful for several groups. For wine importers and producers, it helps determine whether repeated media coverage still creates substantial demand gains or whether the majority of the impact of media coverage can be gained through the initial few reviews. If the first review has the biggest impact and later reviews have significantly less impact, firms may choose to spend resources on getting first-time coverage instead of repeated reference to the same wine. Since expert reviews have already been shown to have economically significant effects on wine demand, understanding whether these effects diminish with repetition is directly valuable for marketing strategy.

For media outlets and reviewers, the answer would indicate whether repeated coverage of the same wines is beneficial to readers and the market, or whether resources would be better spent on new wines instead. If repeated reviews have only small additional effects in demand, it may suggest that wider product coverage may create more value for consumers. It is important in wine markets because expert information has been shown to affect consumer choice (Friberg and Grönqvist (2012)).

For economists, it helps them more broadly understand how information affects demand in markets for experience goods. Studying this question can improve our understanding of consumer behaviour and information efficiency in experience goods markets.

4 Conclusion

This report integrates the main research results of the paper, the discussion on consumer behaviour and market impact, as well as the proposed research questions. It analyses how expert comments affect wine demand. The research results show that expert comments do have a significant impact on demand, especially when they are positive. Such comments usually lead to a significant increase in sales, and this effect can last for a long time. At the same time, the paper adopts a relatively solid empirical method, effectively dealing with endogeneity and other issues, thereby enhancing the reliability of the research results.

From the discussion, it is clear that expert reviews matter mainly because they reduce information problems for consumers. Since wine is an experience good, people cannot fully judge its quality before buying it. Reviews can help consumers better understand the expected effects of the products, thus making it easier for them to make decisions among numerous options. In terms of the market, although there is a certain degree of substitution among different wines, this influence is not very significant. Reviews mainly increase the demand for the wines being reviewed rather than significantly reducing the demand for other wines. Moreover, there is evidence suggesting that reviews can, to some extent, enhance the overall market demand.

The analysis of product differentiation reveals that both quality differences and consumer preferences are of great significance. Positive evaluations lead to an increase in demand, indicating that consumers respond to the perceived quality. However, negative evaluations do not significantly reduce demand, suggesting that there are differentiated preferences among consumers, and some consumers still choose wines that suit their tastes.

Based on these findings, the report raises a new research question: whether repeated assessments have diminishing marginal effects over time. This issue is an extension of the original paper, focusing on whether the mechanism of action will change when consumers are exposed to similar information many times. This helps to further understand whether additional evaluations still have value or if their impact will gradually weaken. Overall, expert evaluations play a crucial role in the formation of the wine market demand. They not only help consumers cope with uncertainty but also influence their decision-making methods, and their impact may extend beyond the product itself.

5 References

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